

Relative Performance Feedback and Redistribution Preferences: Evidence from an Online Experiment*

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Abstract

This paper examines how information about relative performance affects distributive preferences and beliefs about the role of effort versus luck in generating inequality. Using an online experiment, participants complete a code-recognition task and are randomly assigned to one of three feedback conditions: no feedback, told they performed *above* a randomly matched peer, or told they performed *below* a randomly matched peer. They then act as impartial spectators and distribute a fixed sum between two workers whose inequality arose from either differential performance (merit scenario) or a lottery (luck scenario). Participants told they outperformed their peer redistribute approximately \$0.59 less to the lower-earning worker in the merit scenario compared to the no-feedback baseline — a 24 percent reduction from the mean — and score 0.78 points higher on an effort-attribution belief scale (0–10). No significant effect is found for participants told they underperformed, and the luck scenario serves as a placebo with no treatment effects. These results are consistent with self-serving attribution: learning that one outperformed a peer reinforces beliefs that inequality reflects effort, which in turn reduces redistribution demand in merit-relevant contexts.

Keywords: redistribution, relative performance feedback, self-serving attribution, fairness,

*I am grateful to Professor Andrea FM Martinangeli for supervision and guidance. All errors are my own. This experiment builds on the design of Almås et al. (2020) and is inspired by the research question of Cassar and Klein (2019).

online experiment

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1 Introduction

Free-market economies inherently generate unequal outcomes, and while modern democratic societies rely on redistribution to mitigate these disparities, there is persistent disagreement over which inequalities are considered legitimate (Alesina and Giuliano, 2011; Piketty, 2014). A central dimension of this debate concerns the *source* of inequality: outcomes attributed to effort are typically viewed as more deserving than those arising from luck. This intuition is formalised in the accountability framework, which models redistributive preferences as judgments about deservingness and predicts lower redistribution when inequality reflects effort rather than chance (Konow, 2000; Cappelen et al., 2007). Underlying this framework is a well-documented empirical regularity: beliefs about income determinants drive redistribution preferences (Piketty, 1995; Alesina and Angeletos, 2005; Bénabou and Tirole, 2006). People who believe inequality arises from effort tend to favour less redistribution; those who attribute it to luck or circumstance tend to favour more.

To isolate such normative judgments from material self-interest, studies often employ impartial spectator designs in which allocators distribute income between others and receive no payoffs themselves (Almås et al., 2020). While this approach successfully removes monetary incentives, it does not imply psychological neutrality. A growing body of evidence shows that recent experiences of success or failure can shape how individuals evaluate deservingness, even when they subsequently make allocation decisions on behalf of others (Cassar and Klein, 2019; Deffains et al., 2016). In this sense, impartiality is material rather than cognitive, and spectators’ own experiences may influence how they apply the accountability principle.

The stakes of understanding what shapes these beliefs extend to political polarization. Deffains et al. (2016) argue that the experience of success and failure is itself a driver of redistributive polarization through the self-serving bias (SSB): individuals who succeed at a task attribute their success to effort and favour less redistribution, while those who fail attribute failure to circumstance and shift toward more. Because success and wealth are correlated, this mechanism creates a systematic wedge between rich and poor in their perceived causes of inequality. Wealthy individuals, as Deffains et al. (2016) put it, “do not only oppose redistribution because they expect to be net payers, but also because the SSB systematically shifts their fairness principles.” The implication is stark: anything that alters an individual’s sense of their own relative standing could reshape redistribution preferences — not through material self-interest, but through belief updating about whether inequality reflects effort or luck. A natural follow-up question is what shapes those beliefs. Social comparisons are ubiquitous: workers learn their rank in a wage distribution, students receive

class rankings, and online platforms continuously display relative performance metrics. Does learning where one stands in a distribution affect how one interprets the sources of inequality — and through that channel, redistribution preferences?

This paper provides experimental evidence on this mechanism. Participants complete a performance task and are randomly assigned to one of three conditions: (i) *no feedback* about relative standing; (ii) told they performed *above* a randomly matched peer (*above* group); or (iii) told they performed *below* a matched peer (*below* group). Subsequently, all participants act as impartial spectators and decide how to distribute a fixed monetary sum between two workers, in two scenarios: one where inequality arose from differential performance (merit), and one where it arose from a lottery (luck).

The key findings are three. First, participants told they outperformed a peer redistribute significantly less in the merit scenario: \$1.87 on average versus \$2.47 in the no-feedback group, a difference of \$0.59 that is statistically significant at the 5 percent level in OLS regressions with and without controls. Second, the same group scores approximately 0.78 points higher on an effort-attribution scale (0–10), consistent with self-serving belief updating, though this difference is only marginally significant individually ($p \approx 0.086$) while the joint test across groups is significant ($p = 0.028$, ANOVA). Third, the effect is asymmetric: participants told they underperformed do not significantly change their redistribution or belief scores relative to the no-feedback baseline. Taken together, these results suggest that positive performance signals trigger self-serving attribution that translates into reduced redistributive generosity — but that negative signals do not produce a symmetric reverse effect.

These results are consistent with the self-serving attribution account of redistribution developed by Deffains et al. (2016): individuals who perform well relative to others update their beliefs about the role of effort in generating inequality and become less redistributive. Our design extends their framework by using *randomly assigned comparative feedback*, providing clean causal identification and isolating the relative rather than absolute performance channel. One notable difference is that Deffains et al. (2016) find SSB effects on both sides — success reduces redistribution and failure increases it — whereas our evidence is concentrated on the winning side. The absence of a significant response among participants told they underperformed suggests that the SSB in our setting is more asymmetric: learning one beat a peer is enough to trigger meritocratic updating, but learning one lost does not produce a symmetric shift toward redistribution. The absence of any treatment effect in the luck scenario further supports the attribution channel over non-specific mood or attention effects.

The paper most directly relates to Almås et al. (2020), from whose spectator design we

adapt both the merit and luck redistribution tasks, and to Cassar and Klein (2019), whose focus on how performance comparisons shape distributive preferences motivates our research question. Our contribution is to provide clean experimental identification of the direction and asymmetry of this effect using randomly assigned relative feedback.

The rest of the paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the experimental design. Section 4 presents the results. Section 5 concludes.

2 Related Literature

This paper contributes to three strands of literature.

Beliefs about inequality and redistribution. A foundational insight from Piketty (1995) is that beliefs about social mobility and the role of effort versus luck determine preferences for redistribution in equilibrium. Alesina and Angeletos (2005) provide empirical evidence: Americans attribute poverty more to individual failure than Europeans, which partly explains the cross-country gap in redistribution. Bénabou and Tirole (2006) model how self-serving motivated reasoning sustains two equilibria — a “belief in a just world” and a more luck-oriented view — with different steady-state redistribution levels. Our experiment directly tests the mechanism by which an informational shock (relative performance feedback) updates beliefs and behaviour.

Impartial spectator and fairness views. Cappelen et al. (2007) and Cappelen et al. (2013) use spectator designs to classify subjects into fairness types (libertarian, meritocratic, egalitarian) and show that a majority of individuals accept some redistribution away from luck-generated inequality even when indifferent to merit-generated inequality. We adapt their two-scenario design (merit and luck) to test whether third-party redistribution decisions are affected by information about the spectator’s own relative performance — a test of whether the spectator role is truly impartial.

Self-serving attribution and social comparisons. Self-serving attribution refers to the tendency to attribute successes internally (effort) and failures externally (luck or circumstance) (Miller and Ross, 1975; Mezulis et al., 2004). In an economic context, Babcock and Loewenstein (1995) document that parties in bargaining inflate self-assessments of a fair outcome in a self-serving direction. Closer to our design, Deffains et al. (2016) show that individuals who experience unfair treatment become more supportive of redistribution.

Cassar and Klein (2019) show that framing one’s own performance as a failure (relative to a counterfactual) shapes distributive preferences; our design extends this insight by using randomly assigned comparative feedback, allowing clean causal identification. We add to this literature by focusing on the *comparative* and *asymmetric* dimension: the same absolute performance level can produce different attribution and redistribution depending on whether one learns one stands above or below a peer, and the effect appears concentrated on the winning side.

3 Experimental Design

The experiment was conducted online using JATOS and involved two distinct roles: *workers* and *spectators*.

Workers ($n = 2$) completed the code-recognition task under real monetary incentives. They were told in advance that their earnings would depend on their relative performance against each other, and that a third party (a spectator) would subsequently decide whether and how much to redistribute between them. Workers received real monetary payments determined by both their relative task performance and a randomly selected spectator’s redistribution decision.

Spectators ($n = 134$) were recruited through personal networks (friends, family, and colleagues from two workplaces), completed the full experiment online in approximately 7–10 minutes, and received no monetary compensation. Spectators were informed before making their redistribution decisions that the workers were real individuals who had completed the same task and would receive actual monetary payouts determined by one randomly selected spectator’s allocation. The impartial-spectator design thus combines the removal of material self-interest (spectators have no stake in the allocation) with real stakes for the workers, ensuring that decisions carry genuine consequences.

3.1 Part 1: Code-Recognition Task

Participants completed a code-recognition task over five rounds of 45 seconds each. In each round, a target code was displayed at the top of the screen. Participants were shown a grid of codes and had to check as many instances of the target as possible. Scoring: +1 point per correct identification, −1 point per false alarm, 0 points for a miss. The maximum possible score was 200 points; sample means range from 91 to 97 points across treatment groups (Table 1).

Before seeing their results, participants answered three pre-feedback self-report questions:

(i) effort exerted (slider, 0–10); (ii) guessed score (numeric, range -200 to 200); (iii) relative performance prediction (“I would perform better than or the same as the other participant” vs. “I would perform worse”).

3.2 Treatments: Relative Performance Feedback

After the self-report, participants were randomly assigned to one of three feedback conditions.

1. **No feedback.** Participants were informed they could proceed to Part 2 without seeing their score or any comparison information.
2. **Above.** Participants were shown their score and told: “You performed above your randomly matched peer.”
3. **Below.** Participants were shown their score and told: “You performed below your randomly matched peer.”

Each spectator in the relative-feedback group was matched to one of the two workers and shown their own score together with whether they outperformed or underperformed that specific worker. The comparison was thus based on real task scores from a real counterpart, making the feedback both credible and consequential: the worker a spectator was compared to was also the worker whose payoff their redistribution decision would partly determine. Approximately half of the relative-feedback group was told they outperformed their matched worker (above) and half that they underperformed (below).

3.3 Part 2: Distribution Decisions

After the feedback screen, all participants acted as impartial spectators and made two sequential redistribution decisions adapted from Almås et al. (2020).

Decision 1 — Merit Scenario

Participants read that two real participants — the two workers who had previously completed the same task — were identified as Worker A and Worker B. Worker A had received a larger initial payment because they performed better; Worker B received less because they performed worse. Participants were asked to decide how to redistribute a total endowment between the two workers. The decision was presented as a set of discrete options covering the full range from keeping full inequality to reversing it, with an equal split as the midpoint. The primary outcome variable is the amount allocated to the lower-earning Worker B (`redist_merit`, in USD).

Decision 2 — Luck Scenario

After completing a demographics survey (including the effort–luck belief slider), participants saw a second decision. The scenario was identical in structure but attributed the earnings difference to a lottery with no regard for performance. The primary outcome variable is the amount allocated to the lottery loser, Worker D (`redist_luck`, in USD).

3.4 Key Outcome and Mediator Variables

- **Redistribution (merit):** USD allocated to Worker B.
- **Redistribution (luck):** USD allocated to Worker D.
- **Effort–luck belief (`effort_luck`):** Recorded after the merit decision via the question: “To what extent do you think the initial income of participants in the previous task was determined by effort versus luck?” (slider: 0 = 100% luck, 10 = 100% effort). This variable serves as the hypothesised mediator between feedback and redistribution.

3.5 Sample

Spectators were recruited through the researcher’s personal networks: friends, family, and employees at two workplaces. The analysis sample consists of $N = 134$ spectators after excluding non-engagers (zero score and zero clicks) and statistical outliers. This is a convenience sample and is therefore not nationally representative; the main implication for external validity is that findings describe the direction of the self-serving attribution effect under incentivized conditions but may not generalise to other populations in terms of magnitudes. Importantly, treatment assignment was random within this sample, so internal validity is not compromised.

Table 1 presents descriptive statistics by treatment group. Groups are balanced on pre-determined covariates: age ($p = 0.99$), gender ($p = 0.14$), task score ($p = 0.57$), and prior belief about relative performance ($p = 0.43$), confirming that randomisation was successful. The effort–luck belief score differs significantly across groups ($p = 0.028$), which is itself a key outcome; importantly, this measure is elicited *after* the feedback treatment, so it cannot confound treatment assignment.

4 Results

4.1 Effort–Luck Beliefs

Figure 1 (left panel) shows mean effort–luck belief by treatment group. Participants told they outperformed a peer score markedly higher on the effort-attribution scale ($\overline{EL}_{\text{above}} = 7.6$, $SD = 2.3$) relative to the no-feedback group ($\overline{EL}_{\text{nofb}} = 6.8$, $SD = 1.9$) and the below group ($\overline{EL}_{\text{below}} = 6.3$, $SD = 2.6$). A one-way ANOVA rejects equality of group means at the 5 percent level ($F(2, 128) = 3.78$, $p = 0.028$).

Table 2 (columns 1–2) presents OLS regressions of `effort_luck` on treatment dummies, with no-feedback as the baseline. Being told one outperformed a peer increases the effort-attribution score by 0.78 points ($SE = 0.45$, $p = 0.086$), while being told one underperformed decreases it by 0.51 points ($SE = 0.50$, $p = 0.306$). Neither pairwise coefficient reaches the 5 percent significance threshold, but the pattern is consistent with a self-serving attribution mechanism: the feedback most likely to be flattering (above) raises effort attribution, while the feedback that implies underperformance (below) produces little symmetric updating. Controlling for age, gender, and task score leaves both coefficients essentially unchanged (column 2), ruling out demographic confounding. Three participants did not complete the effort–luck slider; the belief regressions use $N = 131$.

4.2 Redistribution in the Merit Scenario

The right panel of Figure 1 shows mean redistribution in the merit scenario. The above group redistributes the least (\$1.87, $SD = \$1.38$), compared to the no-feedback baseline (\$2.47, $SD = \$1.18$) and the below group (\$2.39, $SD = \$1.33$). The one-way ANOVA yields $p = 0.065$, marginally failing the conventional threshold at the omnibus level.

Columns 3–4 of Table 2 report the main OLS regressions. Being told one outperformed reduces redistribution by \$0.59 ($SE = \0.27, $p < 0.05$) relative to the no-feedback group, corresponding to a 24 percent reduction from the no-feedback mean. This estimate is robust to adding demographic and performance controls (column 4: coefficient $-\$0.56$, $SE = \$0.27$, $p < 0.05$). In contrast, the below coefficient is small ($-\$0.08$, $SE = \$0.27$, $p = 0.77$) and statistically indistinguishable from zero, confirming the asymmetric pattern visible in Figure 1.

These results establish a clear causal chain from relative feedback to redistribution in the expected direction, but concentrated on the winning side: participants who learn they outperformed a peer become measurably less redistributive in a merit context, whereas participants who learn they underperformed do not respond significantly.

4.3 Redistribution in the Luck Scenario

As a placebo, we examine redistribution in the luck scenario, where earnings are explicitly attributed to a lottery and performance feedback is normatively irrelevant. If feedback affected redistribution through a non-specific channel — such as mood, engagement, or demand effects — we would expect treatment differences in both scenarios. Table 1 shows that group means in the luck scenario are \$3.16, \$2.70, and \$3.05 for the no-feedback, above, and below groups, respectively, and the one-way ANOVA cannot reject equality ($p = 0.379$). The null result in the luck scenario is consistent with the merit-redistribution effect operating through performance-specific attribution beliefs rather than a general mood or engagement channel.

4.4 Heterogeneity by Surprise

To probe the attribution mechanism further, we classify participants in the relative-feedback group by whether the feedback was surprising. *Positively surprised* participants predicted they would not outperform their peer but were told they did. *Negatively surprised* participants predicted they would outperform but were told they underperformed.

If self-serving attribution drives the result, the redistribution effect should be sharpest among those for whom the positive signal was unexpected — unexpected confirmation of high standing is arguably more diagnostic and self-affirming than expected confirmation. Column 5 of Table 2 reports OLS regressions of merit redistribution on these surprise indicators, estimated within the relative-feedback group ($N = 91$). Relative to confirmed-outcome participants (those whose feedback matched their prior expectation), negatively surprised participants redistribute \$0.53 more (SE = \$0.30, $p = 0.082$), a marginally significant result in the expected direction. Positively surprised participants do not differ significantly from the baseline (\$0.30, SE = \$0.77, $p = 0.70$). We interpret the marginal result for negatively surprised participants as suggestive that unexpected bad news about relative standing may trigger compensatory redistribution, though the relative-group sample is underpowered for this analysis and caution is warranted.

5 Conclusion

This paper provides experimental evidence that relative performance feedback shapes both beliefs about the role of effort in generating inequality and downstream redistribution preferences. Participants told they outperformed a peer attribute more of the task outcome to effort and redistribute \$0.59 less per round in a merit scenario relative to a no-feedback

baseline — a 24 percent reduction. The effect is asymmetric: underperformers do not significantly change either beliefs or redistribution. A luck scenario, where performance feedback is normatively irrelevant by design, yields no treatment effects, suggesting the mechanism is specific to merit-relevant attribution rather than a general engagement or mood effect.

Taken together, these findings support a self-serving attribution account of redistribution preferences: receiving positive comparative feedback leads individuals to update the belief that inequality reflects effort, and this updated belief reduces their willingness to redistribute in settings where merit is salient. The asymmetry — winners become less redistributive, losers do not become more so — mirrors the well-documented asymmetry in self-serving attribution in psychology (Mezulis et al., 2004) and has implications for the design of performance information systems.

These findings have implications for the study of inequality and redistribution in a world where performance comparisons are increasingly salient — on professional platforms, social networks, and in educational settings. The self-serving updating documented here suggests that rising relative visibility of performance rankings could systematically reduce support for redistribution among those who “win” such comparisons, potentially contributing to the polarisation of redistribution preferences.

Future work should (i) increase sample size to improve power, particularly for the heterogeneity and mediation analyses; (ii) test whether the effect persists when participants receive absolute score information alongside relative feedback, to disentangle comparative and informational channels; and (iii) conduct a formal mediation analysis to quantify the share of the redistribution effect that operates through the effort-attribution belief measure.

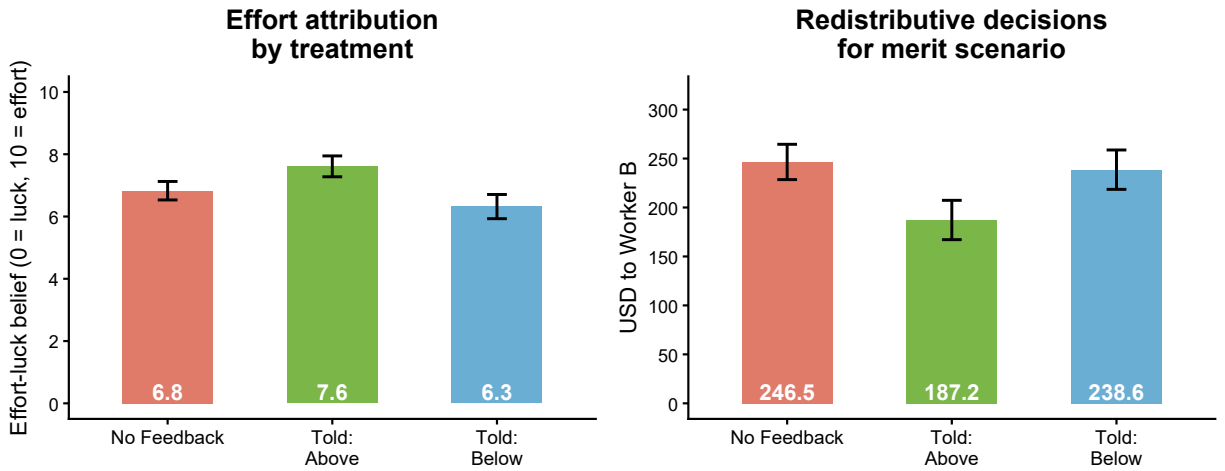
Tables and Figures

Table 1: Descriptive Statistics by Treatment Group

	No Feedback	Told: Above	Told: Below	<i>p</i> -value
Redistribution – merit (\$)	246.51 (118.22)	187.23 (137.71)	238.64 (133.32)	0.065
Redistribution – luck (\$)	315.79 (144.31)	270.27 (156.11)	305.13 (141.33)	0.379
Effort–luck belief (0–10)	6.83 (1.92)	7.61 (2.27)	6.32 (2.55)	0.028
Task score	91.37 (25.51)	96.49 (28.82)	91.25 (26.26)	0.571
Age	34.95 (15.37)	35.28 (13.89)	34.93 (15.02)	0.992
Female (%)	55.81 (50.25)	59.57 (49.61)	75.00 (43.80)	0.141
Predicted better (%)	83.72 (37.35)	89.36 (31.17)	79.55 (40.80)	0.433
<i>N</i>	43	47	44	134

Mean (SD) for continuous variables; mean $\times$ 100 (SD $\times$ 100) for binary. *p*-values from one-way ANOVA (continuous) or chi-squared test (binary). Redistribution in USD.

Figure 1: Effort Attribution and Merit Redistribution by Treatment Group



Notes. Bar heights = group means. Whiskers = ± 1 SE. *N* = 134.

Notes. Left panel: mean effort–luck belief slider (0 = 100% luck; 10 = 100% effort), measured after the merit redistribution decision. Right panel: mean amount allocated (in USD) to the lower-earning worker in the merit scenario. Bar heights correspond to group means; whiskers represent ± 1 standard error. *N* = 134.

Table 2: OLS Regressions: Effort–Luck Belief and Merit Redistribution

	(1)	(2)	(3)	(4)	(5)
Told: Above	0.782 (0.453)	0.785 (0.456)	−59.278* (27.295)	−55.514* (27.387)	
Told: Below	−0.510 (0.495)	−0.540 (0.511)	−7.875 (27.315)	−9.502 (28.953)	
Positively surprised					29.804 (76.668)
Negatively surprised					52.661 (30.034)
Age		0.027 (0.015)		−1.323 (0.952)	
Female		0.148 (0.421)		7.878 (24.362)	
Task score		−0.003 (0.008)		−0.710 (0.524)	
<i>N</i>	131	131	134	134	91
<i>R</i> ²	0.054	0.093	0.041	0.065	0.034
Controls (age, female, score)	No	Yes	No	Yes	No

* $p < 0.05$, ** $p < 0.01$

HC3 robust standard errors in parentheses. Baseline: no feedback (cols 1–4); relative group only (col 5). * $p < 0.05$, ** $p < 0.01$.

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